How far observational insulin sensitivity sits below entered settings depends on how it is measured

Six trial cohorts, 1,679 people, and a gap that ranges from 0.30 to 1.12 according to construction

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Summary

Sensitivity estimated from what glucose did after insulin was given generally comes out below the sensitivity people have entered into their pumps. That was established in 2026 on 138 users of open-source closed-loop systems, where the constant anchored to measured sensitivity was 145 against 355 anchored to the profiles people had tuned, a ratio of 0.41. The same work tested whether dosing to the measured value would be safe and found it would not: people dosing well weaker than their measured sensitivity had more hypoglycaemia rather than less, and restricting the estimate to overnight periods left the bias unchanged.

This paper asks whether that ratio holds on other data. Across 1,679 people in six trial cohorts it does not hold at a single value, and the reason is that the ratio depends on how the estimate is constructed more than on which cohort is measured. Holding the cohort fixed and varying the construction moves it from 0.30 to 1.12. Running the original construction unchanged on the trial cohorts gives 0.72 to 1.12, with the open-loop cohort exceeding its entered settings at six hours.

So the direction of the earlier finding is not universal and the magnitude is not transferable. What is transferable is the caution: an observational sensitivity estimate is a statement about a method as much as about a person, and two defensible methods on the same records differ by more than threefold.

What was already known

The earlier analysis reached three conclusions relevant here. The shape of the relationship between sensitivity and total daily dose is derivable from data and scales approximately as one over the square root of daily dose. The level is not: a short-window empirical estimate swings by a third between blocks and carries almost no memory from one block to the next. And the level that comes out is biased in a specific direction, low, such that dosing to it produces more hypoglycaemia than dosing to a tuned profile.

That last point is the one that matters for interpretation. The gap between measured and entered sensitivity is not evidence that entered settings are wrong. It had already been tested against outcomes and the entered settings won.

What this adds

The cohorts are the Jaeb Center public releases: REPLACE-BG, which ran in 2015 with no algorithm intervening, the Loop observational study, DCLP3, DCLP5 and PEDAP on Control-IQ across ages 2 to adult, and IOBP2 on a bionic pancreas. Ages run from 2 to 82 and daily doses from 7 to 107 U. Insulin on board is reconstructed from the delivery record and an insulin action curve, which is the weaker provenance: the earlier work had the loop's own figure.

The original construction is the fall over four hours divided by insulin on board at the start, taken overnight, with carbohydrate screened by the shape of the trace rather than from a log. Run unchanged on these cohorts it gives:

Cohort	4 h	6 h	Entered	Ratio at 4 h	Ratio at 6 h
REPLACE-BG	42.5	50.4	45.0	0.94	1.12
Loop	45.9	50.7	55.0	0.83	0.92
PEDAP	137.4	144.0	158.8	0.87	0.91
DCLP3	31.2	33.6	41.0	0.76	0.82
DCLP5	54.9	49.4	68.3	0.80	0.72

That is 0.72 to 1.12, not 0.41, and the open-loop cohort exceeds its entered settings once the horizon covers the insulin duration.

Three other constructions were also run, differing in decisions that turn out to matter as much as the cohort does. Whether a correction is counted as a bolus alone or as anything delivered above the programmed basal, which matters because 56% of Loop's correcting insulin and 43% of PEDAP's arrives as raised temporary basal. Whether the denominator is the units given, the insulin action across the window, or insulin on board at the start alone. And whether episodes are required to open with little insulin already on board.

Construction	Cohorts	Ratio
Insulin on board at the start, 4 to 6 h	5	0.72 to 1.12
Bolus corrections, per unit given	5	0.49 to 0.92
Any route, insulin action across the window	5	0.48 to 0.66
Any route, action, low insulin on board at the start	4	0.30 to 0.65
oref archive, loop-recorded insulin on board	1	0.41

The constructions bracket rather than agree, and they do so in an interpretable direction. Dividing by insulin on board at the start ignores anything delivered inside the window, which lowers glucose and enlarges the numerator without appearing below the line, so that construction reads high and reads highest where an algorithm is dosing throughout. Dividing by the action of all above-schedule insulin counts insulin that was covering something else, so it reads low. The honest statement is that the answer lies between them and this data does not place it.

Why the magnitude cannot be pinned here

Three mechanisms produce a low estimate and this design separates none of them.

Glucose falls on its own from a raised level. Overnight and carbohydrate-free, between 150 and 250 mg/dL, it drops a median 21 to 30 mg/dL over four hours with no correction given, so any estimator that removes that fall is measuring something smaller than a settings file encodes, and any estimator that leaves it in is measuring mean reversion.

Insulin is given when glucose is expected to stay up. Within an identical glucose trace the correlation between insulin committed before a window and insulin delivered during it is 0.054 in the open-loop cohort and 0.447 under a do-it-yourself loop, so a controller keeps responding to something the trace does not carry and its dose stays tied to the outcome.

And the insulin action curve is assumed rather than observed. Reconstructed insulin on board agrees with the pump's own record at a median absolute difference of 0.079 U and a correlation of 0.848 across 188 people in REPLACE-BG, which is good but not exact.

A limit on what a simulation can settle

The obvious way to choose between constructions is to run them against records whose true sensitivity is known. That was done and it does not resolve the question. Simulated open-loop records with unlogged carbohydrate return 51.6 against a true 50 for the bolus-and-units-given construction, but only about 22 qualifying episodes per person, and the simulated controller cannot be made to correct through temporary basal at a realistic rate: at any gain that leaves glucose in a plausible range it never delivers half a unit above schedule in thirty minutes. So the constructions that matter for the closed-loop cohorts cannot be tested against a known answer at all.

Reporting that limitation is the point of this section. An estimator validated on the one case that can be simulated should not be presented as validated for the cases that cannot.

What follows

For anyone setting a pump, nothing changes. The entered constants remain the number to use and the earlier outcome work is the reason.

For anyone deriving sensitivity from device data, which includes every dynamic sensitivity equation in current use, the finding is that the quantity being derived is defined by the construction as much as by the person. Two defensible choices of denominator on the same overnight records differ by more than threefold, and the choice that ignores insulin delivered during the window reads highest exactly where an algorithm is delivering it. An equation calibrated against one construction inherits its bias and there is no way to read that bias off the published number.

For anyone reporting such an estimate, the practical recommendation is to publish the denominator in full and the ratio against entered settings alongside the estimate. Every construction here looks defensible in isolation and they span 0.30 to 1.12, so neither the estimate nor the ratio means anything without the construction attached.

The remaining question is why the oref archive returns 0.41 under a construction that returns 0.72 to 1.12 on trial data. That is now a question about those records rather than about method, and one difference is not yet held fixed: the oref figure came from fitting a constant over the square root of daily dose across users, where these are medians of per-person ratios.

Method

Analysis is local, in Python, against the Jaeb releases loaded into PostgreSQL 17.9 with TimescaleDB 2.26.1. The corpus holds 132,467,585 CGM readings, 68,995,020 basal records and 5,154,053 bolus records.

Delivery is taken net of the programmed basal throughout, since basal offsets hepatic glucose output rather than lowering glucose. The programmed rate comes from the recorded schedule in Loop and REPLACE-BG, from the 48 half-hourly rates on the case report form in DCLP3, DCLP5 and PEDAP, and from each person's own median delivery at that half hour in IOBP2, which programmes no basal at all.

Entered settings are the per-person median of what was recorded at the time of dosing or on the case report form. They were checked against the pump's own arithmetic: the sensitivity implied by dividing the glucose excess by the recommended correction agrees with the recorded column at a ratio of 1.000 in Loop.

Code is at https://github.com/tim2000s/dynamic-isf-calculations. The assembled ratios are produced by `inv009/ratio_evidence.py`, the estimators by `inv009/correction_landing.py` and `inv009/correction_routes.py`, and the simulation check by `inv009/audit_estimators.py`.

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